

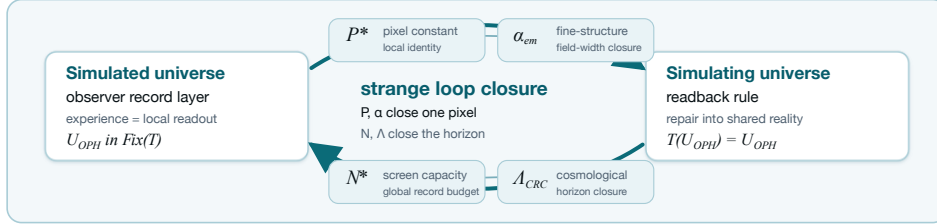
Disclosure Day

A short proof that we experience a simulation

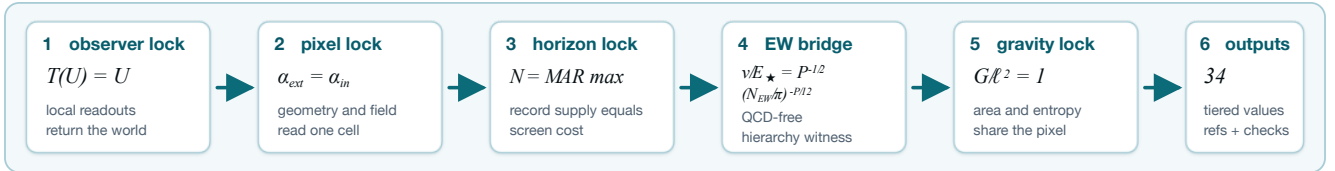
Paper release: r1489 Released: June 15, 2026

Observer Patch Holography does not use “simulation” as a metaphor. It models reality as a finite observer-patch computation whose terminal normal form is experienced as the physical universe [O1,O2,O3]. The carrier chart contains finite patch algebras, overlap ports, records, mismatch scores, repair moves, and checkpoint continuation. The experienced chart is the quotient-normal form read by observers [O1,O2]. OPH makes the strange-loop pattern explicit: simulated records generate the simulating rule, and the simulating rule returns the same simulated records. For most readers, this sounds insane. The test is simple: measured numbers may locate a branch basin; the branch must emit exact values without a target-value ancestor. This note applies that rule to P_* , N_* , gravity, and the table below. The proof chain is rigorous [O1,O2,O3,O4,O5,O6] and supported by early experimental evidence [S12,S13].

Bracketed source keys are load-bearing. When this note names a theorem without printing its certificate, the numbered OPH paper carries that certificate. The direct audit anchors at the end point to the online source lines.



The labels name each value’s job: pixel identity, fine-structure closure, screen capacity, and cosmological horizon closure.



Six locks. OPH’s five axioms are listed in *Observers Are All You Need* [O1]. The reconstruction theorems force exactly two dimensionless branch constants, P_* and N_* , both computed below [O1,O3,O4,O5,O6]. The observer lock says a universe is a fixed point of its own readback [O1,O2]. The pixel lock says a physical pixel cannot remain purely golden; the source-emitted electromagnetic width detunes it [O3,O4,O5,O6]. The horizon lock says a closed universe cannot borrow memory; record supply and screen cost close on N_* [O1,O4]. The hierarchy lock is the clean quantitative witness: P_* , $N_c = 3$, and an interval-certified $\alpha_U(P_*)$ generate the weak-to-UV ratio [O5]. The gravity lock cancels P_* out of Newton coupling [O3,O4]. The output lock is the tiered table at the end [O4,O5,O7,O8,O9].

OPH is hard to dismiss because one observer-closure rule makes subjective experience a local readout [O1], fixes the low-energy electromagnetic width as pixel detuning [O6], fixes the horizon capacity [O4], interval-certifies $\alpha_U(P_*)$ [O5], generates a 10^{-17} -scale weak hierarchy [O5], fixes the EW-refined capacity N_{CRC}^{EW} [O4,O5], cancels P_* out of gravity [O4], and returns the numerical rows.

From those two forced constants, the axioms, and the cited theorems, this note derives the QCD-free electroweak hierarchy witness, the dimensionless gravity theorem, and the SI display path for Newton’s constant. Its SI display is $G_{OPH} = 6.674299995910528 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. CODATA is comparison data [S1]. The compact evidence chain is stricter than the table: architecture, the two closure coordinates, the EW-refined hierarchy bridge, source-audited structural recovery, and explicit falsifiers [O3,O4,O5]. As a caveat, the SI G display is clock-gated by missing QCD/hadronic refinements inside the cesium clock branch [O4,O5,O6]. Together with the table, the cited stack also recovers quantum theory, general relativity, the Standard Model branch, and the string-vacuum selection layer from observer closure [O1,O4,O5,O9]. The chance that the same two closures do all of this by coincidence is astronomically small. That is why the compact claim is simulation closure, not a lucky value of G .

1. Yes, the universe is a simulation

OPH uses “simulation” in the hard mathematical sense. A physical world is an admissible finite patch-carrier whose observer patches expose ports, write records, compare overlaps, repair disagreement, and return a quotient normal form after readback [O1,O3]. The observer-facing universe is that normal form as experienced by its internal observers. OPH adds no separate computer. The simulating chart is the universe’s own finite consistency layer.

The symbol $\mathcal{U}_{OPH}$ denotes the selected OPH universe. The map T is the observer-overlap repair map [O1,O2]. The notation $Fix(T)$ means the states returned unchanged by that map:

$$T(\mathcal{U}_{OPH}) = \mathcal{U}_{OPH}.$$

Read the equation as the simulation claim stripped to its mechanism. The simulated record world enters the simulating readback rule, contradictory branches die, and the returned object is the same record world. That is a self-contained simulation [O1,O2]. A universe with unexplained external constants would fail this test [O1,O3,O4].

The observer lock changes the meaning of experience. An observer in OPH is a finite record boundary with a comparison rule for nearby patches [O1,O2]. Subjective experience is that boundary reading itself from the inside:

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\text{OPH}}), \quad \mathfrak{U}_{\text{OPH}} = T(\{r_i\}_{\text{compat}}).$$

Here r_i is observer patch i 's local readout, $\mathcal{R}_i$ is its readout map, and $\{r_i\}_{\text{compat}}$ is the compatible readout set. The map T repairs those records and returns the shared world. Left equation: subjective experience. Right equation: shared reality. No readout, no record. No overlap, no public reality. Experience makes reality because OPH reality is the stable object made by compatible observer readouts [O1,O2]. No mysticism: records read, compare, repair, return.

2. Two constants from closure

The pixel and horizon locks are where the usual constants disappear. The simulated/simulating balance leaves two dimensionless constants. The first closes the local pixel: one elementary observer cell must look the same when read as simulated record content and as simulating screen rule. The second closes the global screen: the universe must contain exactly the record capacity needed to simulate the universe that contains it. The local constant is certified in [O3,O4,O5,O6], and the global capacity constant is certified in [O1,O4]:

$$P_\star \quad \text{and} \quad N_\star.$$

The symbol $P_\star$ is the local pixel fixed point. It says how much area and shared edge entropy one elementary observer cell carries. It is the simulated/simulating conversion rate for one cell. The symbol $N_\star$ is the global screen-capacity fixed point. It says how many record units fit on the closed observer-facing horizon. It is the simulated/simulating conversion rate for the whole screen. They are branch outputs, cross-locked by the 24-tick hierarchy relation below. A wrong P, N pair is mostly harmless as a numerical guess; as physics, it misses that gate.

Two capacity displays appear below. The rounded de-Sitter value $N_{\text{scr}}^{\text{dS}} \simeq 3.31 \times 10^{122}$ is diagnostic; the exact hierarchy bridge uses $N_{\text{CRC}}^{\text{EW}} = 3.5323546226929906511 \dots \times 10^{122}$ [O4,O5].

Basin hints are not proof inputs

OPH is a timeless causal loop, so approximate measured scales can legitimately select a branch interval I . The proof uses that interval only as a basin label. Let $F : I \rightarrow I$ be an OPH source map built from declared source data $\mathcal{S}$ [O3,O4,O5,O6], with no occurrence of the measured target q_{obs} or an equivalent calibrated proxy. If

$$F(I) \subseteq I, \quad d(F(x), F(y)) \leq q d(x, y), \quad 0 \leq q < 1,$$

Banach's theorem gives one endpoint $x_\star$. The readout $q_\star = R(x_\star)$ is determined by F and R . Inverting a final display proves a coordinate relation. Circularity requires a dependency path from the measured target into F . The measurement labels the basin. Closure supplies the decimal.

The local constant is computed from a target-free readback equation [O3,O4,O5,O6]:

$$P \mapsto \alpha_U(P) \mapsto A_Z(P) \mapsto A_T^{\text{src}}(P), \quad \Gamma_{\text{src}}(P) = \varphi + \frac{\sqrt{\pi}}{A_T^{\text{src}}(P)}.$$

Here P is a candidate identity value for one OPH pixel: the number that converts the cell's screen area into the edge entropy it shares with neighboring cells. The symbols $\alpha_U(P)$, $A_Z(P)$, and $A_T^{\text{src}}(P)$ are the source unification width, the electroweak anchor, and the Ward-projected Thomson endpoint emitted by that same branch. The simulating geometric reading and simulated electromagnetic reading are

$$\alpha_{\text{ext}}(P) = \frac{P - \varphi}{\sqrt{\pi}}, \quad \alpha_{\text{in}}(P) = \frac{1}{A_T^{\text{src}}(P)}, \quad \alpha_{\text{ext}}(P_\star) = \alpha_{\text{in}}(P_\star),$$

The last equality is the pixel lock, equivalently $P_\star = \Gamma_{\text{src}}(P_\star)$. The golden ratio $\varphi = (1 + \sqrt{5})/2$ is the geometric screen balance point, and $\sqrt{\pi}$ converts boundary width into the pixel chart. The low-energy electromagnetic width is read out as

$$\alpha_\star(0) = \frac{1}{A_T^{\text{src}}(P_\star)} = \frac{P_\star - \varphi}{\sqrt{\pi}}.$$

Solving the source equation determines $P_\star$ and $\alpha_\star(0)$ together. Approximate measured α may identify the branch interval; it never defines A_T^{src} or Γ_{src} . At the fixed point one may write

$$P_\star = \varphi + \alpha_\star \sqrt{\pi}$$

as the public endpoint coordinate.

Fine-structure endpoint status

The compact source-only chain excludes the public Thomson endpoint. The source trunk gives $\alpha_{\text{source}}^{-1} = 136.9948351646 \dots$; the public endpoint $\alpha^{-1}(0) = 137.035999177(21)$ needs the Ward-projected hadronic spectral payload and same-scheme bridge [O5,O6].

The selected value is

$$P_\star = 1.630968209403959324879279847782648941 \dots$$

The number is the unique fixed point in the physical pixel interval [O6]. A different decimal in the same branch basin changes the starting guess and leaves the endpoint unchanged. Local self-simulation demands this fine-tuning: the simulated cell and the simulating boundary must read the same cell [O3,O4,O6].

The global constant is computed from the screen-capacity readback equation [O1,O4]:

$$N_\star = \mathcal{C}(N_\star).$$

Here N is a candidate total capacity for the closed observer screen: the number of record units the universe can carry in OPH units. $\mathcal{C}$ is the closed-screen readback rule. The simulated reading counts the closed-screen normal forms supported at capacity N . The simulating reading charges the screen for the N slots it claims. A self-contained simulation has no external memory bank, so the available records and the cost of the screen must close on the same capacity. The global loop says the simulated universe contains the capacity of the simulating screen that contains it [O1,O4].

The equivalent count-density display is

$$N_\star = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N] \simeq 3.31 \times 10^{122}.$$

The set Ω_N^{sc} contains the closed-screen normal forms at capacity N . The term $\log |\Omega_N^{\text{sc}}|$ is the simulated-side count, while the subtraction of N is the simulating-side budget charge. The selected $N_\star$ is the capacity where record availability and record cost agree under OPH readback. The operator MAR is Axiom 5 of OPH [O1]: Minimal Admissible Realization. In the core papers, Axiom 5 says that the realized branch is the lexicographically minimal admissible representative under the branch complexity vector $C(\mathfrak{S})$. In this screen-capacity display, the same selector chooses the minimal admissible arithmetic representative of the maximizing branch. The count-density problem records the same selection: record availability and record cost close on the same value.

Why forced means forced

$P_\star$ and $N_\star$ instantiate the basin test above. For $P_\star$, the source pixel map Γ_{src} stays inside the physical pixel interval and shrinks candidates [O3,O4,O5,O6]. For $N_\star$, closed-screen counting plus MAR selects the minimal admissible capacity branch [O1,O4]. The decimals are forced because each branch has one certified endpoint.

The QCD-free hierarchy witness

The load-bearing quantitative witness is the electroweak hierarchy. It is deliberately QCD-free: using the public Thomson endpoint, hadron masses, or the cesium clock would import the Ward-projected hadronic payload as an assumption. The source map fixes $\alpha_U(P_\star) = 0.041124336195630495$ in the interval $I_U = [0.041123336195630494, 0.041125336195630496]$, with the Krawczyk image strictly inside I_U [O5].

The factor 12 is screen topology. On the triangulated S^2 screen, the locally flat triangular vacuum is six-valent. Define the curvature charge $q_v = 6 - \deg(v)$. Since $V - E + F = 2$ and $3F = 2E$, one has $E = 3V - 6$, hence

$$\sum_v q_v = 6V - \sum_v \deg(v) = 6V - 2E = 12.$$

Refinement-stable MaxEnt spreads this total charge into twelve unit fivefold defects: at fixed $\sum q_v = 12$, the convex local defect cost $\sum q_v^2$ is minimized by unit charges, while negative defects require extra positive charge. Edge-center completion turns each defect collar into a central port [O3]. With no marked point, the twelve equal ports sit on the maximal finite rotation orbit A_5/C_5 , whose size is $60/5 = 12$. Therefore the global screen depth $D = \log(N/\pi)$ is sampled locally as $D/12$. The local cell entropy is $\ell_{\text{cell}} = P_\star/4$, and the electroweak projection coefficient is $\beta_{\text{EW}} = N_c + 1 = 4$. Therefore

$$\Gamma_{\text{EW}} = \beta_{\text{EW}} \ell_{\text{cell}} \frac{\log(N/\pi)}{12} = \frac{P_\star}{12} \log(N/\pi).$$

With $E_{\text{cell}} = E_\star/\sqrt{P_\star}$, this gives

$$\frac{v}{E_{\text{cell}}} = \left(\frac{N_{\text{CRC}}^{\text{EW}}}{\pi} \right)^{-P_\star/12}, \quad \frac{v}{E_\star} = P_\star^{-1/2} \left(\frac{N_{\text{CRC}}^{\text{EW}}}{\pi} \right)^{-P_\star/12} = 2.0199803239725553 \times 10^{-17}.$$

Equating this exponent with the source transmutation certificate $\exp[-\pi/(2\alpha_U)]$ gives

$$\alpha_U(P_\star)^{-1} = \frac{P_\star}{6\pi} \log(N_{\text{CRC}}^{\text{EW}}/\pi), \quad \mathcal{B}_{\text{EW}}(P_\star, N_{\text{CRC}}^{\text{EW}}) = 0.$$

No measured G , Λ , M_W , M_Z , Higgs mass, Planck area, low-energy electromagnetic endpoint, or QCD hadronic endpoint is an ancestor of this map [O5]. Ordinary particle physics treats the Higgs scale as a tuning problem. In OPH the weak scale is a branch output with zero naturality defect:

$$\epsilon_H = 0, \quad \epsilon_H \in [0, 0].$$

Zero means there is no cancellation to arrange [O4,O5].

Empirical evidence. Alex Osika's PIXEL lane tests the same local pixel closure in a bench protocol: sweep P , drive $\text{SU}(2)/\text{SU}(3)$ heat-kernel weights, and verify $\bar{\ell}_{\text{SU}(2)}(t_2) + \bar{\ell}_{\text{SU}(3)}(t_3) = P/4$ [S12]. IBM quantum-cloud runs test independent OPH fingerprints $Z_5 \rightarrow \varphi^2$ and corrected $S_3 \rightarrow 2$ [S13].

3. Gravity without units

Gravity usually enters physics as a dimensional constant. OPH gets it from a pixel identity before any unit system appears. The proof is short because $P_\star$ is an identity tag, not a tuning knob. OPH identifies cell area and shared-edge entropy as two

readings of the same pixel, then Newton's coupling becomes the conversion between them [O3,O4]:

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

Here G_{geom} is Newton's coupling in OPH geometric units, a_{cell} is the area of one observer cell, and $\bar{\ell}_{\text{shared}}$ is the shared edge-entropy weight of that cell. The equation makes gravity the conversion factor between an area count and the edge entropy an observer patch shares with its neighbors. The factor 4 is the quarter-area normalization from horizon entropy and the OPH edge-entropy branch.

The selected pixel gives

$$a_{\text{cell}} = P_{\star} \ell_{\star}^2, \quad \bar{\ell}_{\text{shared}} = \frac{P_{\star}}{4}.$$

The symbol $\ell_{\star}$ is the OPH length quantum, so $\ell_{\star}^2$ is the OPH area quantum. The first equation assigns physical area to one cell; the second assigns the same cell its shared entropy weight. Both use $P_{\star}$, because $P_{\star}$ is exactly the dimensionless pixel ratio.

Substitution gives the dimensionless gravity result:

$$\frac{G_{\text{geom}}}{\ell_{\star}^2} = \frac{a_{\text{cell}}/\ell_{\star}^2}{4\bar{\ell}_{\text{shared}}} = \frac{P_{\star}}{4(P_{\star}/4)} = 1.$$

The pixel branch fixes every symbol in this line. Divide the geometric Newton coupling by the OPH area quantum and the ratio is one. The same $P_{\star}$ measures cell area and shared edge entropy, so the pixel value cancels instead of becoming an adjustable gravitational constant.

Thus

$$G_{\text{geom}} = \ell_{\star}^2.$$

The local gravity theorem is one line [O3,O4]. The SI value needs a separate no- G clock certificate [O4].

4. Gravity as a number in SI units

The SI decimal is the clock translation of the unit-free theorem. The source burden is the dimensionless cesium gap [O4]:

$$\varepsilon_{\text{Cs}}^{\text{src}} = \mathcal{H}_{\text{Cs}}^{\text{src}}(\mathcal{D}_{\text{Cs}}^{\text{src}}), \quad \mathcal{D}_{\text{Cs}}^{\text{src}} = \left(\alpha_{\star}, \frac{m_e}{E_{\star}}, \frac{\Lambda_{\text{QCD}}}{E_{\star}}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\text{atom}}^{\text{Cs}} \right).$$

Here $\mathcal{H}_{\text{Cs}}^{\text{src}}$ is the source-side cesium hyperfine readout and $\mathcal{D}_{\text{Cs}}^{\text{src}}$ is its dimensionless packet: electromagnetic width, electron and QCD ratios, quark Yukawa records, ^{133}Cs nuclear data, and atomic cesium binding data. Forbidden ancestors are measured G , ℓ_P , m_P , t_P , or any gravity-calibrated proxy.

The displayed clock record is [O4]:

$$\varepsilon_{\text{Cs}} = 3.1139305134160128239 \dots \times 10^{-33}, \quad \gamma_{\star} = 4.9559743365484194637 \dots \times 10^{-34}.$$

ε_{Cs} is the clock gap. $\gamma_{\star}$ is the same record in the length chart.

Missing QCD refinements

As a caveat, the local theorem $G_{\text{geom}} = \ell_{\star}^2$ is QCD-free. The SI decimal depends on ε_{Cs} , whose endpoint record includes hadronic transport through nuclei and cesium hyperfine structure [O4,O5,O6]. Full finite-lattice QCD source simulation is infeasible on conventional von Neumann hardware at the required resolution, so the compact SI row assumes the QCD/hadronic refinement. This does not affect the dimensionless gravity theorem or the QCD-free hierarchy witness. If ε_{Cs} is back-solved from measured G , the SI row is a display checksum. It counts as source-only evidence only when the no- G clock hierarchy emits the cesium gap. OMEGA optical fixed-point compute hardware is built for this source-record class [S14].

The display step uses exact SI constants [S3,S4]:

$$\gamma_{\star} = \frac{\ell_{\star} \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}.$$

Here $\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ s}^{-1}$ and c are exact SI definitions. The 2π converts angular phase to cycles.

Solving for the length quantum gives

$$\ell_{\star} = \frac{c \varepsilon_{\text{Cs}}}{2\pi \nu_{\text{Cs}}}.$$

This turns the dimensionless clock gap into a length.

The geometric-to-SI conversion is

$$G_{\text{SI}} = \frac{c^3}{\hbar} G_{\text{geom}} = \frac{c^3 \ell_{\star}^2}{\hbar}.$$

This unit-chart display would be tautological if $\ell_{\star}$ were imported from measured G . The OPH source burden is exactly the no- G cesium gap above [O4].

Combining the last two equations with $G_{\text{geom}} = \ell_{\star}^2$ gives

$$G_{\text{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\text{Cs}}^2} \varepsilon_{\text{Cs}}^2.$$

The powers are unit bookkeeping: $\ell_\star^2$ contributes $c^2 \varepsilon_{\text{Cs}}^2 / (4\pi^2 \nu_{\text{Cs}}^2)$, and area-to-SI conversion contributes $c^3 / \hbar$. Substitution gives

$$G_{\text{OPH}} = 6.674299995910528 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

The comparison with laboratory G uses [S1] and is external to the derivation. The fixed-point equations above take their inputs from closure records alone [O3,O4,O5,O6].

5. Other values predicted by OPH

One QCD-free hierarchy witness and one pixel cancellation give the short route to G . OPH is a theory of everything, so the same two closure constants organize far more than gravity [O1,O4,O5,O7,O8,O9]. Source-only rows require a target-free map and no backend-gated empirical payload; [C] marks public code.

Falsification rule

OPH cannot promote a row by post-hoc repair. A compact-proof row fails if its declared map contains a target-value ancestor, if its interval certificate fails, if its public evidence bundle is missing, or if an exact verifier rejects the claimed output. The hierarchy row also fails if $N_{\text{CRC}}^{\text{EW}}$ is back-solved from the weak scale or silently replaced by the rounded 3.31×10^{122} de-Sitter display. Hardware claims require hashes, raw readouts, controls, and verifier receipts [O3,S12,S13].

Lane	OPH-derived value	Comparison or interpretation	Tier	Refs
Simulation substrate	$\mathfrak{U}_{\text{OPH}} \in \text{Fix}(T)$	The universe is the stable fixed point of observer-overlap repair.	structural	[O1,O2,C5]
Local pixel	$P_\star = 1.6309682094039593 \dots$	Unique local cell-area and edge-entropy fixed point.	derived	[S12,O3,O4,O5,O6,C1]
Local cell entropy	$\bar{\ell}_{\text{shared}} = P_\star/4 = 0.407742052350989825 \dots$	Shared edge entropy assigned to one observer cell by the same fixed point.	derived	[O3,O4]
Global capacity	$N_\star \simeq 3.31 \times 10^{122}$	Late de-Sitter closed-screen diagnostic display on the selected branch.	diagnostic	[O1,O4]
Screen sieve and repair count	$\sum_v q_v = 12, \quad m_{\text{rep}} = 24$	Euler-forced ports plus product-adjoint repair rhythm.	derived	[O3,O5,C3]
EW bridge capacity	$N_{\text{CRC}}^{\text{EW}} = \pi \exp \left[\frac{6\pi}{P_\star \alpha_U(P_\star)} \right]$	Unique source-side capacity satisfying the exact EW bridge residual: $3.53235462269 \dots \times 10^{122}$.	derived	[O4,O5,C3]
Hierarchy resonance	$v/E_{\text{cell}} = (N_{\text{CRC}}^{\text{EW}}/\pi)^{-P_\star/12} = g'_{\text{EW}} ^{4P_\star}$	Weak scale as the local projection of the global repair law on the EW bridge.	derived	[O4,O5,C3]
Higgs naturality	$\epsilon_H = 0, \quad \epsilon_H \in [0, 0]$	OPH naturally solves the hierarchy row: the selected source-to-Higgs normal form has zero RG/coarse-graining defect.	exact	[O4,O5,C3]
	$\varepsilon_{\text{Cs}} = 3.11393051342 \dots \times 10^{-33}$			
Clock bridge	$\gamma_\star = 4.95597433655 \dots \times 10^{-34}$	The OPH length quantum expressed in the cesium-clock unit chart.	clock-gated	[O4]
	$B_\star = 3.60787391468 \dots \times 10^{70} \text{ m}^{-2}$			
	$\ell_\star^2 = 2.61228030237 \dots \times 10^{-70} \text{ m}^2$			
Area quantum and cell area	$a_{\text{cell}} = 4.26054612722 \dots \times 10^{-70} \text{ m}^2$	The OPH area unit and physical area of one screen pixel.	derived	[O3,O4]
	$G_{\text{geom}} = \ell_\star^2$			
Newton coupling	$G_{\text{SI}} = 6.674299995910528 \times 10^{-11}$	One OPH pixel cancellation gives the geometric coupling; CODATA 2022 lists $G = 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.	gated	[S1,O3,O4,C2]
Cosmological constant	$\Lambda_{\text{CRC}} \approx 1.09 \times 10^{-52} \text{ m}^{-2}$	Planck 2018 late-universe Λ CDM comparison at displayed precision.	hit	[S5,O1,O4]
Static-patch rate	$H_{\text{dS}} \approx 55.759940256 \text{ km s}^{-1} \text{ Mpc}^{-1}$	De Sitter static-patch display, distinct from the fitted local H_0 .	derived	[O1,O4]
Hubble benchmark	$H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Planck 2018 reports $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$.	0 σ	[S5,O1,O4]
Lorentz group	$\text{Conf}^+(S^2) \cong \text{SO}^+(3, 1)$	The support-visible screen branch reconstructs Lorentz kinematics.	structural	[O4]
Causal speed	$c = 299792458 \text{ m s}^{-1}$	Exact SI definition used for the Lorentz speed display.	exact	[S3,O4]
Einstein branch	Jacobson-type local Einstein equation	Local horizon thermodynamics and observer overlap select the Einstein equation branch.	structural	[O3,O4]
Gauge quotient	$\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y / \mathbb{Z}_6$	The Standard Model gauge quotient drops out of the selected compact gauge branch.	exact	[S6,O4,O5]
Hypercharge, color, generations	one-generation hypercharge lattice $N_c = 3, \quad N_g = 3$ $m_\gamma = 0$	Charge lattice, color count, and family count are branch readouts.	exact	[S6,O4,O5]
Protected carriers	$m_\gamma = 0$	Photon, gluon, and graviton masslessness are protected by the selected gauge and diffeomorphism branches.	structural	[S6,O4,O5]
Gauge proton decay	$m_{\text{grav}} = 0$ $\tau_{\text{p}}^{(\text{gauge})} = \infty$	The selected gauge quotient has no simple-GUT gauge-boson proton-decay channel.	allowed by limits	[S6,O5]
Fine-structure source trunk	$\alpha^{-1}_{\text{source}} = 136.994835164621649 \dots$	Source-side electromagnetic trunk before endpoint transport.	source	[O5,O6,C1]
Fine-structure endpoint	$\alpha^{-1}(0) = 137.035999177(21)$ $\alpha(0) = 0.00729735256433 \dots$ $\alpha_2(m_Z) = 0.03377843630219015$	CODATA gives $\alpha^{-1} = 137.035999177(21)$.	empirical	[S2,O5,O6,C1]
Electroweak source	$\alpha_Y(m_Z) = 0.010131601067241624$	Weak and hypercharge couplings on the selected electroweak branch.	declared	[S6,O5,C2]
Electroweak scale	$v = 246.76711732749683 \text{ GeV}$ $M_W = 80.377 \text{ GeV}$	Derived weak scale in the selected convention.	derived	[S6,O5,C2]
W/Z validation pair	$M_Z = 91.18797809193725 \text{ GeV}$ $m_H = 125.1995304097179 \text{ GeV}$	PDG 2024 gives $M_W = 80.3692 \pm 0.0133 \text{ GeV}$ and $M_Z = 91.1876 \pm 0.0021 \text{ GeV}$.	0.18 σ	[S6,O5,C2]
Higgs/top branch	$m_t = 172.3523532883115 \text{ GeV}$ $m_e = 0.00051099895 \text{ GeV}$	PDG 2024 gives $m_H = 125.20 \pm 0.11 \text{ GeV}$ and direct $m_t = 172.57 \pm 0.29 \text{ GeV}$.	-0.004σ -0.75σ	[S6,S7,O5,C2]
Charged leptons	$m_\mu = 0.1056583755 \text{ GeV}$ $m_\tau = 1.7769324651340912$ $u = 0.00216, \quad d = 0.00470$	PDG lepton summary gives the displayed central values.	$\leq 0.03\sigma$	[S8,O5,C2]
Quark sextet	$s = 0.0935, \quad c = 1.273$ $b = 4.183, \quad t = 172.3523532883115$ 0.017454720257976796	PDG quark summary gives the displayed light/heavy quark central values in GeV.	0 σ shown	[S7,O5,C2]
Neutrino masses	$0.019481987935919015 \text{ eV}$ 0.05307522145074924	Absolute individual masses are OPH branch outputs.	target	[S10,O5,C2]
Neutrino sum	$\sum m_\nu = 0.09001192964464505 \text{ eV}$	Planck 2018 plus BAO gives $\sum m_\nu < 0.12 \text{ eV}$ at 95% CL.	below limit	[S5,S11,O5,C2]
Neutrino mixing display	$\theta_{12} = 34.2259^\circ, \quad \theta_{23} = 49.7228^\circ$ $\theta_{13} = 8.68636^\circ, \quad \delta = 305.581^\circ$ $\alpha_{21} = 153.618518^\circ$	NuFIT 6.0 gives global-fit profiles; the row is profile-comparison data.	profile	[S9,O5,C2]
Majorana phases	$\alpha_{31} = 257.003241^\circ$ $\Omega_{\Lambda, \text{OPH}} = 0.684423332$	Absolute-phase neutrino observables are OPH branch outputs.	target	[S11,O5,C2]
Compressed matter budget	$\Omega_A = 0.264114401$ $\Omega_m = 0.315484968$	Planck 2018 gives $\Omega_m = 0.315 \pm 0.007$ and $\Omega_\Lambda \simeq 0.685 \pm 0.007$.	$\Omega_m : 0.07\sigma$ $\Omega_\Lambda : -0.08\sigma$	[S5,O1,O4,O7,C4]
Dark-sector coefficient	$\lambda_c = \exp(-P_\star/24) = 0.934300639489386 \dots$	Finite-thickness dark-sector susceptibility normalization.	derived	[O7,O8,C4]

Source keys. **Data:** [S1]G; [S2] α^{-1} ; [S3]c; [S4] $\hbar$; [S5]Planck; [S6]PDG gauge/Higgs; [S7]PDG quarks; [S8]PDG leptons; [S9]NuFIT; [S10]KATRIN; [S11]PDG neutrinos; [S12]PIXEL; [S13]IBM; [S14]OMEGA. **OPH:** [O1]Observers; [O2]Consensus; [O3]Screen; [O4]Rel/SM; [O5]Zoo; [O6]Fine; [O7]Dark; [O8]Chi-Nu; [O9]String. **Code:** [C1]P; [C2]particles; [C3]hierarchy; [C4]dark; [C5]consensus. **Proof anchors:** axioms [O1:L296–305]; confluence [O2:L529–545]; patch carrier [O3:L155–166]; P [O6:L895–922]; N [O4:L5195–5237]; no- G clock [O4:L292–303]; G [O4:L5698–5727]; α_U [O5:L1538–1559]; EW bridge [O4:L416–442]; hierarchy bridge [O5:L1738–1787]; EW capacity [O5:L1825–1844]; source-only α [O5:L1458–1477]; endpoint boundary [O6:L1018–1024]; tiers [O4:L5388–5400].